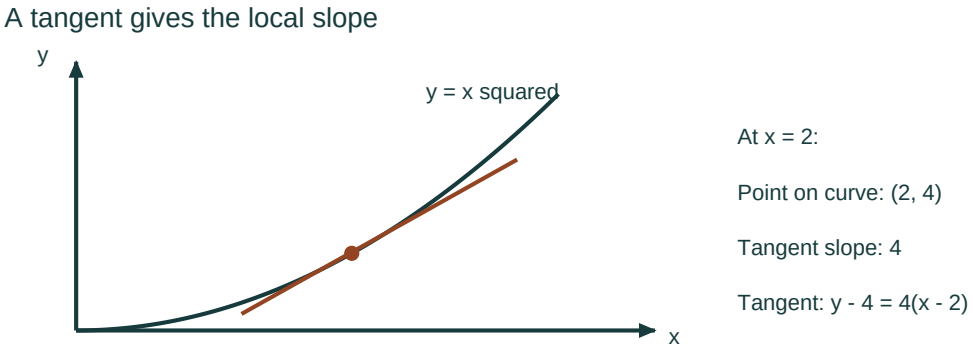
04

# Calculus I

A field guide to understanding, working and checking.




MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

### How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

### Your study route

1. Limits, continuity and the derivative definition
2. Basic derivatives and tangent lines
3. Product and quotient rules
4. Chain rule and trigonometric functions
5. Implicit differentiation
6. Logarithmic, exponential and inverse-trig derivatives
7. L Hopital and indeterminate forms
8. Increasing, decreasing and concavity
9. Graphing, asymptotes and absolute extrema
10. Optimization and mean-value theorems
11. Antiderivatives and substitution
12. Definite integrals, signed area and the fundamental theorem

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**LESSON 01 / 12**

## Limits, continuity and the derivative definition

### After this lesson, you can

- Distinguish a limit from a value at the point.
- Find a polynomial derivative from the difference quotient.

Start with: Expanding a square; factoring and cancelling a nonzero common factor.

### THEORY

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

### RULE

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$h$ small nonzero input increment before the limit (same unit as x)

$f'(x)$ instantaneous rate of change (output unit per input unit)

### A method that earns marks

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.
3. Factor out h and cancel only while h is nonzero.
4. Take the limit h approaching zero in the simplified expression.

### WORKED EXAMPLE

For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$. Letting h approach zero gives derivative 2x.

1. For f(x) = $x^2$, the incremented value is $(x+h)^2 = x^2 + 2xh + h^2$.
2. Subtracting f(x) leaves $2xh + h^2$.
3. Dividing by nonzero h gives 2x + h.
4. As h approaches zero, the secant slope approaches 2x, so f′(x) = 2x.

**Common mistake**
Zero over zero is an indeterminate form, not a final answer of zero.

Exam habit: Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

### SELF-CHECK

Why should h not be set to zero before simplifying the difference quotient?

Answer: It would produce division by zero; the limit concerns nonzero h approaching zero.

LESSON 02 / 12

# Basic derivatives and tangent lines

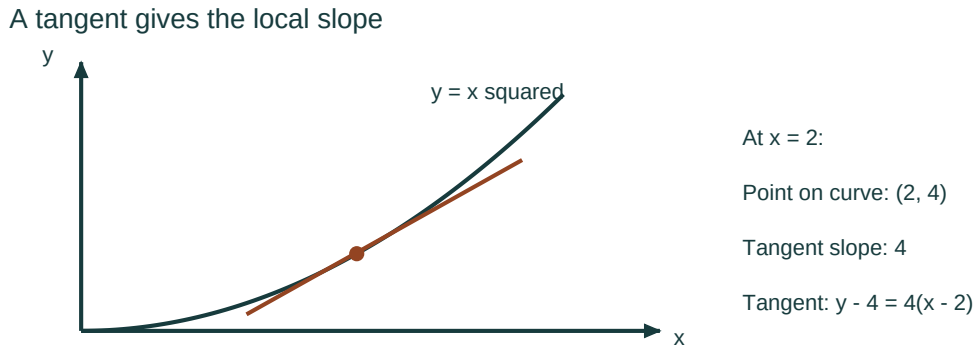
### After this lesson, you can

- Differentiate sums and powers correctly.
- Build a tangent line from the original point and derivative slope.

Start with: Power rule; substituting an x value; point-slope form of a line.

## THEORY

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.




The tangent to $y = x^2$ at (2,4) has slope 4. The axes use different display scales; slope is calculated from coordinate values, not from the visual angle.

## RULE

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

$a$ input coordinate of tangent point (input unit)

$f(a)$ output coordinate of tangent point (output unit)

$f'(a)$ tangent slope (output unit per input unit)

### A method that earns marks

1. Differentiate the original function term by term.

2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

**WORKED EXAMPLE**

For $f(x) = x^2 + 1$ at $x = 2$, the point is (2,5) and slope is 4. The tangent is $y - 5 = 4(x - 2)$, or $y = 4x - 3$.

1. For f(x) = $x^2$ + 1, f′(x) = 2x because the constant contributes zero.
2. At x = 2 the slope is 4, while the original function gives y = 5.
3. The tangent must therefore pass through (2,5) with slope 4.
4. Its equation is y - 5 = 4(x-2), which expands to y = 4x - 3.

**Common mistake**
Do not use $f'(a)$ as the y-coordinate.

Exam habit: Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

**SELF-CHECK**

Does substituting x = 2 into f′ give the y coordinate of the tangent point?

Answer: No. It gives the slope; use f(2) for the y coordinate.

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**LESSON 03 / 12**

## Product and quotient rules

**After this lesson, you can**

- Differentiate a product using both changing factors.
- Preserve term order and the squared denominator in the quotient rule.

Start with: Basic polynomial and sine derivatives; multiplication and fractions.

**THEORY**

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

**RULE**

$$(uv)' = u'v + uv'$$
$$(u/v)' = (u'v - uv')/v^2$$

$u, v$ the two factor functions (problem dependent)

$u', v'$ their derivatives with respect to x (corresponding output per input unit)

### A method that earns marks

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

**WORKED EXAMPLE**

For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$. Both terms are needed.

1. For y = $x^2$ sin x, choose u = $x^2$ and v = sin x.
2. Their derivatives are u′ = 2x and v′ = cos x.
3. The product rule keeps each factor once unchanged: u′v + uv′.
4. Thus y′ = 2x sin x + $x^2$ cos x; multiplying the two derivatives alone would miss both contributions.

**Common mistake**
The quotient denominator is the square of the original denominator.

Exam habit: Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

**SELF-CHECK**

For y = (x+1)/(x-1), what is the derivative for $x \neq 1$?

Answer: -2/$(x-1)^2$: the numerator is (x-1) - (x+1) = -2.

LESSON 04 / 12

# Chain rule and trigonometric functions

### After this lesson, you can

- Identify inner and outer operations in a composite function.
- Include the inner derivative in trigonometric and power derivatives.

Start with: Power rule; sine and cosine derivatives; radians.

#### THEORY

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

#### RULE

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$u = g(x)$ inner expression (problem dependent)

$g'(x)$ inner derivative (inner-expression unit per input unit)

### A method that earns marks

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.
4. Check the sign for cosine derivatives and use radians for trigonometric rules.

#### WORKED EXAMPLE

For $y = (3x+1)^4$, derivative = $4(3x+1)^3 \cdot 3 = 12(3x+1)^3$.

1. In y = $(3x+1)^4$, the inner expression is 3x+1 and the outer operation is raising to the fourth power.
2. The outer derivative gives $4(3x+1)^3$.
3. The inner derivative is 3, so it must multiply the outer result.

4. Combining the constants gives $y' = 12(3x+1)^3$.

**Common mistake**
Forgetting the inner derivative is the most common lost mark.

Exam habit: Before finishing, circle the inner expression and verify that its derivative appears as a factor.

**SELF-CHECK**

What is the derivative of $\cos(x^2)$?

Answer: $-2x \sin(x^2)$, combining the cosine derivative with the inner derivative 2x.

### Four more trigonometric derivatives

The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

- Use radians. Here $u$ is a function of $x$, and $u'$ means its derivative.
- Apply the outside rule, then multiply by the inside derivative.
- Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.

$$\frac{d}{dx}\tan u = \sec^2 u\, u'$$
$$\frac{d}{dx}\cot u = -\csc^2 u\, u'$$
$$\frac{d}{dx}\sec u = \sec u \tan u\, u'$$
$$\frac{d}{dx}\csc u = -\csc u \cot u\, u'$$

Differentiate $y = \cot(5x)$.

1. The inside is $u = 5x$, so $u' = 5$.
2. The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

$y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

LESSON 05 / 12

# Implicit differentiation

### After this lesson, you can

- Differentiate y terms while treating y as a function of x.
- Collect and isolate all terms containing y′.

Start with: Chain and product rules; algebraic rearrangement.

#### THEORY

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces $y'$. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

#### RULE

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

$y'$ dy/dx, the local slope of the curve (y unit per x unit)

### A method that earns marks

1. Differentiate both sides with respect to x.
2. Attach y′ when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y′ to one side.
4. Factor out y′, divide by its coefficient and state where that division is valid.

#### WORKED EXAMPLE

From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$. Thus $y' = -\frac{x}{y}$ wherever y is nonzero.

1. The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value.
2. Differentiating $x^2$ gives 2x; differentiating $y^2$ gives 2y y′ by the chain rule.
3. The constant 25 differentiates to zero, so 2x + 2y y′ = 0.
4. Rearranging gives y′ = -x/y wherever $y \neq 0$; division by y requires this restriction.

**Common mistake**
Do not treat y as a constant or omit the product rule on xy.

Exam habit: For an xy term write y + xy′, not simply y or xy′; both factors depend on x.

**SELF-CHECK**

Why does the derivative of $y^2$ contain y′?

Answer: Because y depends on x; the chain rule multiplies 2y by dy/dx.

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**LESSON 06 / 12**

## Logarithmic, exponential and inverse-trig derivatives

### After this lesson, you can

- Differentiate exponential and logarithmic composites.
- Use logarithmic differentiation for a positive variable base and exponent.

Start with: Chain rule; natural logarithms; product rule; positive-domain restrictions.

**THEORY**

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

**RULE**

$$(e^u)' = e^u u'$$
$$(\ln|u|)' = u'/u$$
$$(\arctan x)' = 1/(1+x^2)$$

$e$ base of the natural exponential (dimensionless)

$\ln u$ natural logarithm of a positive dimensionless argument (dimensionless)

### A method that earns marks

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.

4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

**WORKED EXAMPLE**

For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$. For y=$e^{3x}$, derivative is $3e^{3x}$.

1. For ln(x$^2$+1), the inner derivative is 2x and the inner expression is always positive.
2. The logarithmic chain rule therefore gives 2x/(x$^2$+1).
3. For the separate function e^(3x), the exponential is retained unchanged.
4. Multiplication by the inner derivative 3 gives 3e^(3x).

**Common mistake**
The derivative of ln u is u prime/u, not 1/ln u.

Exam habit: The power rule nx^(n-1) assumes a constant exponent; x^x needs another method.

**SELF-CHECK**

For x > 0, what is d(x^x)/dx?

Answer: x^x(ln x + 1). Take ln y = x ln x, then y′/y = ln x + 1.

### Other exponential, logarithm and inverse-trig rules

The supplied final includes $3^x$ and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

- The base $a$ is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
- For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
- Find the inside derivative $u'$ first. Inverse sine means $\arcsin$; it is different from $1/\sin$.

$$\frac{d}{dx}a^u = a^u \ln(a)\, u'$$
$$\frac{d}{dx}\log_a u = \frac{u'}{u \ln a}$$
$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$$
$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$$

Differentiate $3^x$ and $\log_3 x$.

1. In both cases the inside expression is $u = x$, so $u' = 1$.
2. For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

$(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

LESSON 07 / 12

# L Hopital and indeterminate forms

### After this lesson, you can

- Check for a permitted indeterminate quotient before using L Hopital.
- Evaluate a simple limit by differentiating numerator and denominator.

Start with: Limits; exponential and trigonometric derivatives; chain rule.

### THEORY

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

### RULE

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

$0/0$ indeterminate limiting form, not an arithmetic value (not a measured quantity)

$\infty/\infty$ indeterminate quotient form (not a measured quantity)

### When this applies

The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

### A method that earns marks

1. Substitute the limiting input to identify the quotient form.
2. Confirm 0/0 or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

**WORKED EXAMPLE**

As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$. The derivative quotient is $\frac{e^x}{1}$, giving limit 1.

1. As x approaches zero, e^x - 1 approaches zero and x approaches zero.
2. The quotient therefore has the required 0/0 form.
3. Differentiate numerator and denominator separately to obtain e^x/1.
4. Its limit is e^0 = 1, so the original limit is 1.

**Common mistake**
Do not use the rule on a quotient whose substitution gives an ordinary finite value.

Exam habit: Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

**SELF-CHECK**

What is the limit of sin(3x)/x as x approaches zero?

Answer: 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

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**LESSON 08 / 12**

## Increasing, decreasing and concavity

**After this lesson, you can**

- Use a first-derivative sign chart to find monotonic intervals.
- Distinguish a stationary point from a proven extremum or inflection.

Start with: Factoring quadratics; sign tests; first and second derivatives.

**THEORY**

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

**RULE**

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

$f'$ first derivative, controlling increase or decrease (output per input unit)

$f''$ second derivative, controlling concavity (output per input unit squared)

### A method that earns marks

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.
4. Use f″ signs separately for concavity and require a concavity change for an inflection point.

#### WORKED EXAMPLE

For $f = x^3 - 3x$, derivative is $3(x - 1)(x + 1)$. The function increases outside $[-1, 1]$ and decreases between -1 and 1.

1. For f(x) = $x^3$ - 3x, f′ = $3x^2$ - 3 = 3(x-1)(x+1).
2. The critical numbers are -1 and 1.
3. Testing x = -2, 0 and 2 gives derivative signs positive, negative and positive.
4. The function increases on (-∞,-1) and (1,∞), and decreases on (-1,1); the sign changes identify a local maximum at -1 and minimum at 1.

**Common mistake**
A derivative equal to zero does not automatically mean a maximum.

Exam habit: Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

#### SELF-CHECK

Is f″(c) = 0 alone enough to prove an inflection point?

Answer: No. Concavity must change across c.

LESSON 09 / 12

# Graphing, asymptotes and absolute extrema

### After this lesson, you can

- Find absolute extrema by comparing all candidates on a closed interval.
- Distinguish a removable hole from a vertical asymptote.

Start with: Domain restrictions; critical numbers; evaluating a function at endpoints.

#### THEORY

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

#### RULE

$$\text{Candidates} = \text{endpoints plus interior critical numbers}$$

$[a, b]$ closed interval including both endpoints (input unit)

$f(c)$ function value at a candidate point (output unit)

### A method that earns marks

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.
4. Compare the values and report each extremum value together with where it occurs.

#### WORKED EXAMPLE

For $x^2$ on $[-1, 2]$, candidates are -1,0,2. Values are 1,0,4, so the absolute minimum is 0 and maximum is 4.

1. For f(x) = $x^2$ on [-1,2], the derivative 2x is zero at x = 0.
2. The candidate inputs are the two endpoints -1 and 2 and the interior critical number 0.
3. The original function values are f(-1)=1, f(0)=0 and f(2)=4.
4. The smallest is 0 at x=0; the largest is 4 at x=2.

**Common mistake**
Do not omit endpoints or compare derivative values instead of function values.

Exam habit: Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

**SELF-CHECK**

Can an endpoint be the absolute maximum even when its derivative is not zero?

Answer: Yes. Absolute extrema on a closed interval must include endpoint checks.

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**LESSON 10 / 12**

## Optimization and mean-value theorems

### After this lesson, you can

- Translate a simple optimization constraint into one variable.
- Check mean-value-theorem assumptions and solve for the interior input.

Start with: Area and perimeter; derivatives; continuity and differentiability.

**THEORY**

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

**RULE**

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

$x$ one rectangle side length (length unit)

$A$ rectangle area (square length unit)

$c$ interior input guaranteed by the mean-value theorem (input unit)

### A method that earns marks

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.

4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

**WORKED EXAMPLE**

A rectangle with perimeter 20 has area $x(10-x)$. Its derivative is $10-2x$, so maximum area occurs at $x=5$: a square of area 25.

1. Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x.
2. Area becomes A(x) = x(10-x) with 0 < x < 10 for a nondegenerate rectangle.
3. Its derivative is 10 - 2x, which vanishes at x = 5.
4. The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

**Common mistake**
A candidate outside the physical domain is not a valid optimization answer.

Exam habit: A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

**SELF-CHECK**

For f(x)=$x^2$ on [1,3], what c satisfies the mean-value theorem?

Answer: c=2: the average slope is (9-1)/(3-1)=4 and f′(c)=2c=4.

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**LESSON 11 / 12**

## Antiderivatives and substitution

**After this lesson, you can**

- Choose a substitution whose differential matches the integrand.
- Return the antiderivative to x and include the integration constant.

Start with: Chain rule; power integration; constant factors.

**THEORY**

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

**RULE**

$$\int x^n\,dx = x^{n+1}/(n+1) + C\ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

$u$ chosen inner expression (problem dependent)

$du$ differential of the inner expression (same unit as u)

$C$ arbitrary additive integration constant (antiderivative unit)

### A method that earns marks

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.
4. Substitute back to x, add C and differentiate once to check.

**WORKED EXAMPLE**

For the integral of $2x(x^2+1)^3$, set $u = x^2 + 1$ and $du = 2x\,dx$. Answer is $\frac{(x^2+1)^4}{4} + C$.

1. For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$.
2. Then $du = 2x\,dx$, so the complete integral becomes the integral of $u^3$ du.
3. The antiderivative is $u^4/4 + C$.
4. Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

**Common mistake**
The power rule exception is n=-1; do not divide by zero.

Exam habit: Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

**SELF-CHECK**

What is the integral of $6x(x^2+2)^2$ dx?

Answer: $(x^2+2)^3 + C$. With $u=x^2+2$, the integral becomes 3 times the integral of $u^2$ du.

### Integrate sine, cosine and an exponential

These short rules reverse the chain rule. A sine integral appears in the supplied final.

- Here $a$ and $b$ are constants and $a \neq 0$. Trigonometric angles use radians.

- Divide by the coefficient $a$; differentiation would multiply by it.
- Add $C$ for an indefinite integral. Differentiate your answer to check it.

$$\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + C$$

$$\int \cos(ax+b)\,dx = \frac{\sin(ax+b)}{a} + C$$

$$\int e^{ax+b}\,dx = \frac{e^{ax+b}}{a} + C$$

Find $\int \sin(3x)\,dx$.

1. The sine antiderivative is negative cosine.
2. The inside derivative is 3, so divide by 3.
3. Check: $\frac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

$$-\frac{1}{3}\cos(3x) + C.$$

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LESSON 12 / 12

## Definite integrals, signed area and the fundamental theorem

### After this lesson, you can

- Distinguish signed integral from total geometrical area.
- Apply the fundamental theorem, including a variable-upper-limit chain factor.

Start with: Antiderivatives; positive and negative regions; substituting bounds.

### THEORY

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

### RULE

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

$a, b$ lower and upper integration bounds (input unit)

$F$ an antiderivative of f (f unit times input unit)

$t$ dummy integration variable (input unit)

### A method that earns marks

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

**WORKED EXAMPLE**

The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1. These answer different questions.

1. For y = x from -1 to 1, the region below the axis contributes negative signed area.
2. An antiderivative is $x^2/2$; evaluation at 1 and -1 gives 1/2 - 1/2 = 0.
3. Geometrically each half is a triangle with area 1/2.
4. Total geometrical area is 1/2 + 1/2 = 1, while the signed integral remains zero.

**Common mistake**
Never keep x-limits on an integral that has been fully changed to u.

Exam habit: For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

**SELF-CHECK**

Does a zero definite integral prove the function is zero throughout the interval?

Answer: No. Positive and negative contributions can cancel.

### Read a Riemann sum as rectangles

The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

- Split the interval from $a$ to $b$ into $n$ equal widths.
- Choose a sample point $x_i^*$ in each small interval. Its function value is that rectangle's signed height.
- Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
- Below-axis heights contribute negative signed area.

$$\Delta x = \frac{b-a}{n}$$

rectangle sum $= \sum_{i=1}^{n} f(x_i^*)\,\Delta x$

Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

1. Each width is $\Delta x = (1-0)/2 = 1/2$.
2. The right endpoints are $1/2$ and $1$, giving heights $1/2$ and $1$.
3. The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Estimate: $3/4$. Exact integral: $1/2$. The right-end estimate is too large because the function increases.

## Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

# Calculus I / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

### Part I: Choose one answer

1. The derivative of $x^2$ at $x = 3$ is:

   A. 3

   B. 6

   C. 9

   D. 0

2. The derivative of a constant 7 is:

   A. 7

   B. 1

   C. x

   D. 0

3. The derivative of $xe^x$ is:

   A. $e^x(1 + x)$

   B. 0

   C. $e^x$

   D. $xe^x$

4. The derivative of $\sin(5x)$ is:

   A. $-5\sin(5x)$

   B. $\sin(5)$

   C. $\cos(5x)$

   D. $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at (3,4) is:

   A. 0

   B. $\frac{3}{4}$

   C. $-\frac{3}{4}$

   D. $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

   A. $\arctan(1)$

B. $\frac{1}{1+x^2}$

C. $\frac{1}{\cos^2 x}$

D. $-\frac{1}{1+x^2}$

**Part II: Show your reasoning**

1. Use the definition to find the derivative of f(x)=3$x^2$.

2. Find the tangent to f(x)=x cubed at $x = 2$.

3. Differentiate y=(x+1)/(x-1), for x not equal to 1.

4. Differentiate y=cos($x^2$).

5. Find dy/dx for xy+y squared=6.

6. Differentiate y=x to the power x for x greater than zero.

# Calculus I / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

### Part I: Choose one answer

1. Before applying L Hopital directly, the quotient should have form:

   A. $\frac{0}{0}$ or $\frac{\infty}{\infty}$

   B. $\frac{1}{1}$ only

   C. $\frac{0}{5}$ only

   D. Any fraction

2. If $f'$ is negative on an interval, f is:

   A. Decreasing

   B. Constant

   C. Always concave down

   D. Increasing

3. To find absolute extrema on a closed interval, check:

   A. Only where $f'$ is zero

   B. Only endpoints

   C. Endpoints and interior critical numbers

   D. Only where f is zero

4. Rolle theorem additionally requires:

   A. An infinite interval

   B. Equal endpoint function values

   C. A discontinuity

   D. A nonzero derivative everywhere

5. An indefinite integral must generally include:

   A. Only a numerical answer

   B. A fixed upper limit

   C. No variable

   D. An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

   A. 8

B. -4

C. 0

D. 4

**Part II: Show your reasoning**

1. Evaluate the limit of sin(3x)/x as x approaches zero.

2. For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

3. Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

4. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

5. Evaluate the integral of 6x($x^2$+2) squared with respect to x.

6. Evaluate the integral from 0 to 1 of 2x($x^2$+1) dx.

# Calculus I / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

### Part I: MCQs

1. **B.** The derivative is 2x, so its value at 3 is 6.
2. **D.** A constant function has zero rate of change.
3. **A.** The product rule differentiates each factor in turn.
4. **D.** The inner derivative contributes a factor of 5.
5. **C.** Implicit differentiation gives slope $-\frac{x}{y}$.
6. **B.** This is the standard inverse-tangent derivative.

### Part II: Written solutions

1. [3(x+h) squared -3$x^2$]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.
2. f(2)=8; derivative=3$x^2$; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.
3. Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.
4. Outer derivative is -sin($x^2$); inner derivative is 2x. Multiply to obtain -2x sin($x^2$). Credit both layers, sign and product.
5. Differentiate: y+x $y'$+2y $y'$=0. Collect: (x+2y)$y'$=-y. Hence $y'$=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.
6. Take logarithms: ln y=x ln x. Differentiate: $y'$/y=ln x+1. Multiply by y: $y'$=x to the x times (ln x+1). Four marks for these stages.

# Calculus I / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

### Part I: MCQs

1. **A.** The rule concerns specified indeterminate quotient forms.
2. **A.** Negative slope means the function decreases as x increases.
3. **C.** The largest or smallest value may occur at either kind of candidate.
4. **B.** Equal endpoint heights make the average slope zero.
5. **D.** Differentiation loses constants, so all antiderivatives differ by a constant on an interval.
6. **C.** The odd integrand has equal and opposite signed areas.

### Part II: Written solutions

1. The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.
2. Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.
3. Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.
4. The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.
5. Set u=$x^2$+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = ($x^2$+2) cubed+C. Credit substitution, factor, integration and constant.
6. Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

## Source and coverage notes

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.